

class 17: part 2

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Population Scale Analysis

Q13. Read this file into R and determine the sample size for each genotype and their corresponding median expression levels for each of these genotypes.

How many samples do we have?

```
expr <- read.table("expressiondata.txt")
head(expr)
```

	sample	geno	exp
1	HG00367	A/G	28.96038
2	NA20768	A/G	20.24449
3	HG00361	A/A	31.32628
4	HG00135	A/A	34.11169
5	NA18870	G/G	18.25141
6	NA11993	A/A	32.89721

```
nrow(expr)
```

```
[1] 462
```

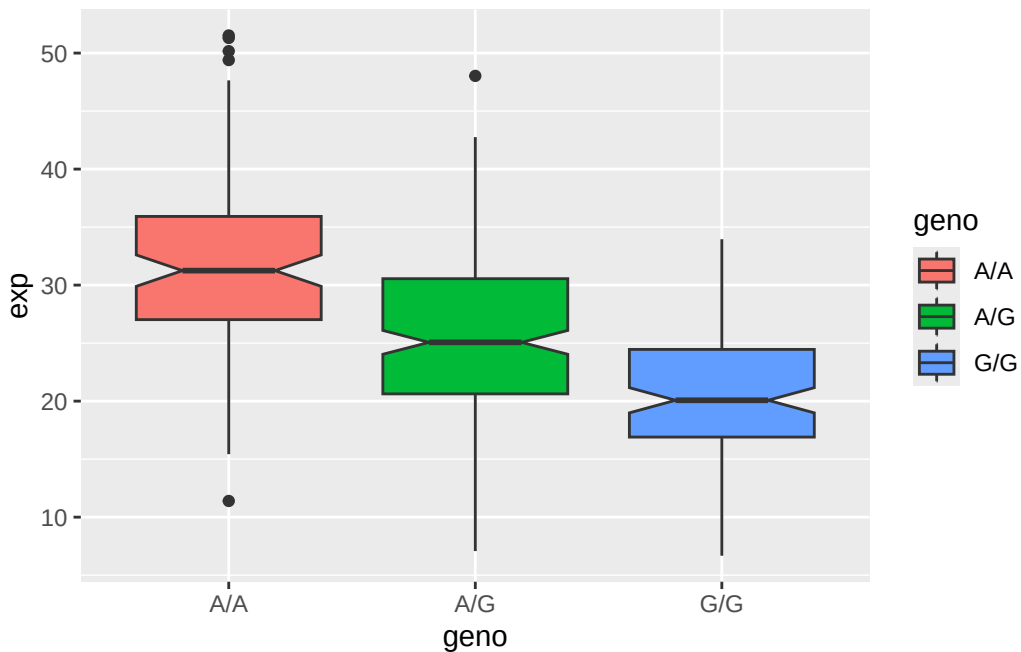
```
table(expr$geno)
```

```
A/A A/G G/G
108 233 121
```

```
library(ggplot2)
```

Lets make a boxplot

```
ggplot(expr) + aes(x=geno, y=exp, fill=geno) +  
  geom_boxplot(notch=T)
```



Q14: Generate a boxplot with a box per genotype, what could you infer from the relative expression value between A/A and G/G displayed in this plot? Does the SNP effect the expression of ORMDL3?

G/G at this location results in reduced expression of this gene; A/A and A/G have higher expression than G/G. The SNP does influence the expression in an additive manner; each copy of the G allele reduces ORMDL3 expression. Therefore, the SNP does affect expression of ORMDL3, with G/G individuals showing reduced expression of the gene.